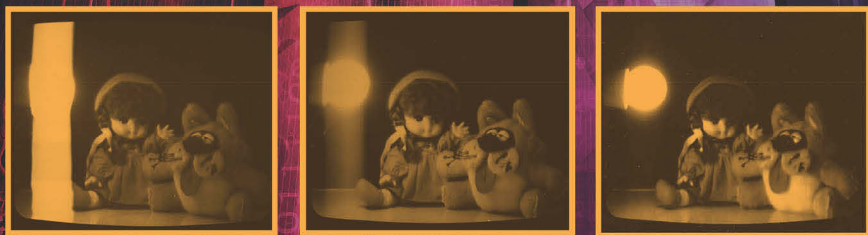
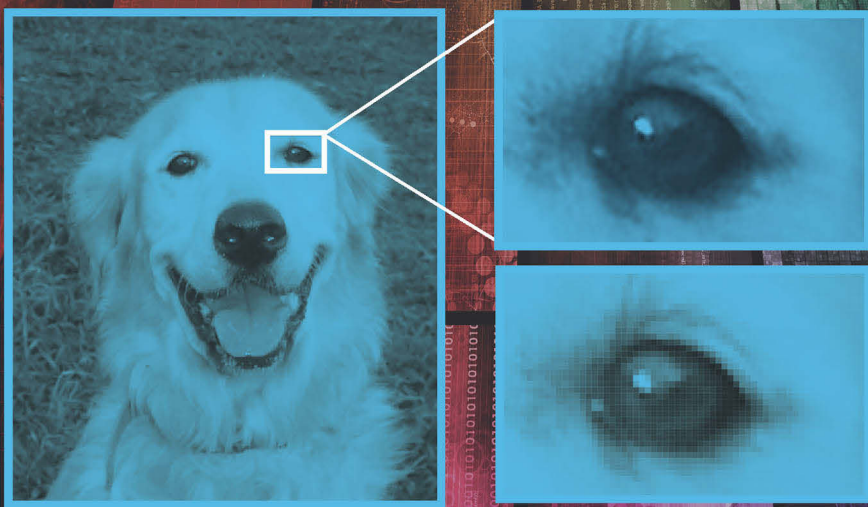


Essential Principles OF Image Sensors

TAKAO KURODA



CRC Press
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Preface to English Edition

It is my greatest pleasure that this book concerning image sensors will be available for people to read all over the world.

The Japanese edition of this book was published in December 2012. Its aim was to explain the “indispensable functions and ideal situation of image sensors” by marshaling the “factors defining image information quality” and understanding the “structure of image information.” It also discussed the history of image sensors and the way they tried to respond to market demands by using the best combinations of the available technology in each era. The book also attempted to investigate future directions of this field.

Although this book was started only 5 months after the publication of the Japanese edition, there was significant progress in this short time, and these advances are described in this edition. While the content of this book is “essential principles,” as the title shows, it also deals with the “state of the art” regarding the surrounding issues.

This edition has come to be published thanks to Professor Dr. Jun Ohta of the Nara Institute of Science and Technology, who kindly introduced me to CRC Press/Taylor & Francis. The author genuinely expresses his gratitude. Thanks are extended to the relevant staff of CRC Press/Taylor & Francis for their hearty support.

I hope that this book enables readers to survey the whole world of image sensor technology through an understanding of the essential principles.

Finally, I gratefully acknowledge the support of my family. My wife, Toshiko, established and has been maintaining the circumstances to enable me to devote myself wholeheartedly to my job for 35 years and checked the English throughout the book. There is little doubt that this book, including the Japanese edition, would not have been born without her support. I thank my two daughters, Chiaki and Chihiro, who encouraged me and helped with the English and some of the artwork from overseas. I also thank my dog, Simba, who was a treasured member of our family and appears in many of the sample images throughout this book. She healed and heartened me with her gentle and friendly nature, charming and cute smile, elegant appearance, and amazing intelligence.

Takao Kuroda
Ibaraki, Osaka, Japan



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Preface to Japanese Edition

More than thirty years have passed since I entered the field of image sensor technology. During this time, my main focus has been how image sensors should be understood. During these years, I have finally achieved a solution. In a sense, this book is also a reminder to sort out my remaining tasks. Thus, I ask readers for permission to present many of my own unique ways of understanding, explanations, structures of logics, and sometimes biases.

At university, I studied physics, which favors a unified perspective, and I believe that to understand is to be able to call up an image of the phenomenon in the mind. To realize my belief, I have tried to visualize physical phenomena by using my original figures to help readers' intuitive understanding.

Chapter 1 presents the structure of image information and the role of sensors to make it clear what "imaging" is, as the basis of this book. In Chapter 2, semiconductor components and circuit components, necessary for sensors, will be dealt with. In Chapter 3, the noise in sensors will be described. In Chapter 4, the scanning mode will be dealt with to conclude the preparatory stages.

Chapter 5 contains a detailed explanation of the principles, pixel technologies, and progress of CCD, MOS, and CMOS sensors. Their current situations will be compared at the end of the chapter. Chapter 6 contains an explanation of how information other than light intensity is obtained. In Chapter 7, the techniques that have improved information quality of each factor constructing the image will be described. In Chapter 8, the elements concerning the information quality of images, not of single sensors but of whole imaging systems, will be described.

In publishing this book, I am indebted to many people in this field, including my elders and seniors, some of whom have already passed away. And in my office, I was so happy to have respectable bosses, nice seniors, colleagues, and juniors, including a few people who showed examples of how not to behave. I would especially like to sincerely thank Dr. Kenju Horii, who was my first direct boss in the office and guided me from the very beginning. My work would not have continued to today without meeting Dr. Horii. I found my true vocation by providence. I also would like to express special thanks to Mr. Yoshiyuki Matsunaga, who was seated beside me during my last years in the office. We had many discussions as engineers of image sensors and also as humans. I learned much from his words and behavior.

As I wrote this book at home, I am very grateful for the constant support of my family, wife, two daughters, and dog, Simba, who often appears in this book in the sample images and was so gentle, generous, and smart. She always cheered me up, but sadly she passed away while I was writing the final chapter of the book.

In publishing this book, I specially acknowledge Professor Dr. Takayuki Hamamoto of Tokyo University of Science for his kind efforts and the relevant members of the Corona Publishing Company. I would also like to thank the authors whose figures from papers and websites I have referred to. As for the relevant literature on the fundamental techniques, I have tried to refer to the most original works or those published earliest.

If this book is of help in guiding readers who want to understand the world of image sensors but are at a loss, as I was in younger days, I will be most grateful.

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Author

Takao Kuroda received his bachelor and master's degrees in material physics and PhD from Osaka University, Osaka, Japan, in 1972, 1974, and 1978, respectively.

He joined Panasonic Corporation in Osaka in 1978 and started research and development of image sensors, beginning with charge-coupled devices, at the Panasonic Electronic Laboratory. He extended the field of development to shift-register addressing-type sensors in addition to charge-coupled devices. In 1981, a sensor model he designed, a solid-state single-chip color camera system, was employed for the cockpit cameras of Japan Airlines to show views at takeoff and landing to passengers for the first time. In 1985 he was in charge of evaluation in the development of the first mass-produced charged-couple device model for Panasonic's single-chip consumer color camcorder. In 1986 he developed the first charged-couple device model for consumer camcorders that featured an electronic shutter function. He also developed a very-low-smear-level charged-couple device with a new photodiode structure, which was reported at the International Solid-State Circuits Conference in 1986.

In 1987, he moved to Panasonic's Kyoto Research Laboratory to establish a charged-couple device developmental regime and processing line. In 1989, he happened to encounter high-energy ion implantation technology, which he instantly recognized as having remarkable potential for performance improvement of image sensors, especially for charged-couple devices. He had a clear view of their absolute necessity, as present conditions prove. After confirmation of the performance improvement of a prototype charged-couple device, he installed the equipment in the laboratory. He was the manager of the elemental technology development group.

He was developing image sensors considering the essentiality of imaging and future integration with computers and communication. As a result, he and his group developed a technique that could remarkably increase the maximum charge quantity transferred in charged-couple devices, as reported at the International Solid-State Circuits Conference in 1996. This brought about significant performance advances in all Panasonic charged-couple devices, some of which were used in camcorders by a competitor charged-couple device company.

As he had recognized the potentiality of combining complementary metal-oxide semiconductor sensors with the pixel technology of charged-couple devices, he embarked upon this project for the first time at Panasonic in 1996, despite much dissent.

In 1998, he moved to the image sensor business unit to establish a business strategy, while addressing the issues of the charged-couple device mass-production line, which had shifted to the Tonami factory, and developing a complementary metal-oxide semiconductor sensor. He was in charge of reinforcement of the development system and intellectual property from 2001 to 2005.

Initially he retired from Panasonic in December 2005, but at the request of the company in January 2006, he took up a post as advisor to the image sensor business unit to establish a technology strategy based on his views on technology and business.

He retired completely from Panasonic in 2011 to write a book on image sensor technology. The Japanese edition was published in 2012 by Corona Publishing Co., Ltd.

Dr. Kuroda was a member of the subcommittee on imagers, microelectromechanical systems, and medical and display devices for the International Solid-State Circuits Conference from 2006 to 2008. He is a fellow of the Institute of Image Information and Television Engineers of Japan.

He holds 70 Japan patents and 15 U.S. patents.

Task of Imaging and Role of Image Sensors

Various kinds of instruments such as digital still cameras (DSCs), camera phones, and camcorders allow us to enjoy personal images. Moreover, broadcasting cameras, which provide high-definition images, are indispensable to the television industry. Not only are there cameras for personal enjoyment, but there are also cameras for other applications such as for automobiles and security systems, and endoscopes for medical use. Cameras are not only used for visible light, but they are also used for thermography, which visualizes thermal distribution by infrared imaging; there are also cameras for ultraviolet and x-ray imaging. In addition, there are cameras that capture very-high-speed phenomena and cameras that obtain highly accurate color information. Furthermore, there are cameras whose images are used not by the human eye but by machines, such as cameras for automated driving and machine vision, which judge information obtained from images. As just described, various kinds of cameras are utilized in a very wide range of fields.

Why are there so many kinds of imaging systems as typified by cameras? The reason is to obtain images with adequate image quality for the purpose of each imaging system. In each imaging system, the role of each image sensor is to pick up image information of high enough quality for that system.

In this chapter, the factors that determine image information are confirmed. Then, the structure of image sensor output and image information are set out. Unless explicitly stated otherwise, the explanations are based on “almost all image sensors” (see Section 1.2.3). Concerning the terms used in this book, *image information* is used for image information in a broader sense, *optical image information* is used for information contained in optical images, and *image signal* is used for image sensor output obtained from optical images.

1.1 Factors Constructing Image Information

What is image information made up of? For the sake of simplicity, let us initially focus on a monochrome still picture. There is a concentration distribution of black and white in two-dimensional space in a monochrome still picture. The concentration indicates the light intensity, which is brighter at lower concentrations. That is, the concentration is the light intensity distribution at each position in two-dimensional space. Therefore, an image is constructed using the information on space (position) and the intensity of the light at that position.

Let us now consider color still images. Since information on the wavelength of light must be added, color image information is formed by light intensity, space, and wavelength. Moreover, in the case of color moving images, time information when light reaches the image should be added. Thus, the image information is composed of four factors: light intensity, space (position), wavelength, and time.¹

Among these factors, space has two dimensions, however wavelength information is often replaced and approximated by the primary colors red, green, and blue, as will be shown later; therefore, color can be considered three dimensional and time has one dimension. Thus, as shown in Figure 1.1, a distribution of the four factors in seven dimensions, that is, a set of each of the coordinate points of light intensity, space, wavelength, and time constructs the image information.

The indexes that indicate the level of information quality are accuracy and range, as shown in Figure 1.2. While the accuracy of value is the resolution capability, which means the signal-to-noise ratio (SNR), the range is the extent of the signal information that the imaging system can pick up. Information captured with high accuracy and over a wide range is high-quality image information. In the case of light intensity, for example, the information quality is decided by the level of SNR and the dynamic range, which describes the maximum and minimum measurable light intensities that the imaging system can capture, as shown in Figure 1.3.

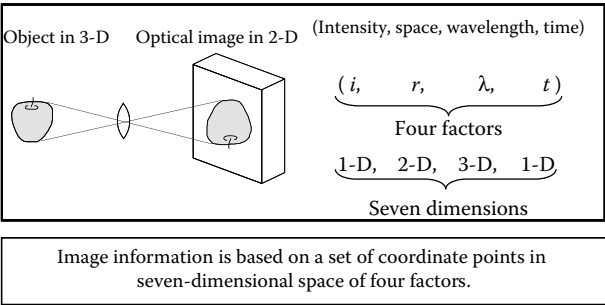


FIGURE 1.1
Structure of optical image information.

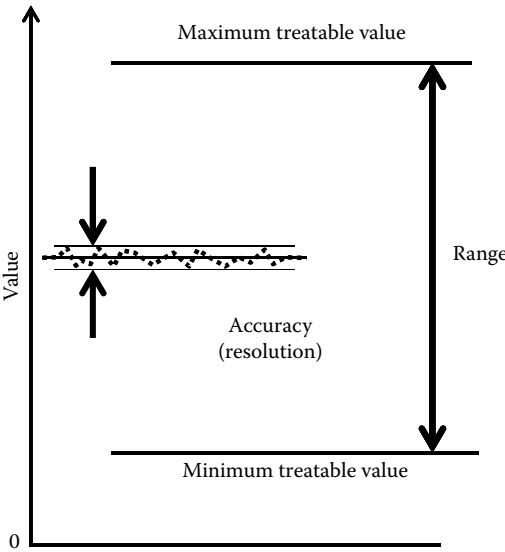


FIGURE 1.2
Quality of image information: Accuracy and range.

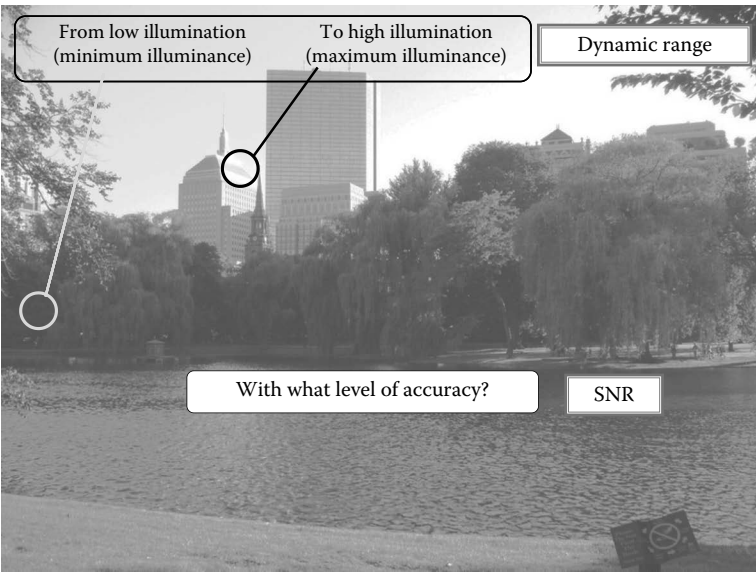


FIGURE 1.3 (See color insert)
Example of the quality of light intensity information.

The accuracy and range of the four factors are shown in Table 1.1. The accuracy of space information is space resolution, that of wavelength is color reproducibility, and that of time is time resolution. The ranges of space,* wavelength, and time are the capturing space range, the color gamut or wavelength range, and the storage time range,† respectively. Apart from special applications and the dynamic range of intensity, these ranges are rarely a problem.

In addition to intensity and wavelength, polarization and phase are among the light conditions. Although these are not discussed in this book, some examples deal with this information, such as sensors to obtain the polarization distribution,² which concerns the surface conditions of materials and textured surfaces, and sensors³ and systems⁴ to obtain depth and range information using phase difference.⁵ There is a sensor⁶ that gets range information from the angle information of incident light. Since there is a strong need for information on range and depth, progress in this field is expected.

TABLE 1.1
Accuracy and Range of Four Factors

Factor	Accuracy (Resolution)	Range
Intensity	SNR (sensitivity)	Dynamic range
Space	Space resolution	Space range
Wavelength	Color reproducibility	Color gamut, wavelength range
Time	Time resolution	Storage time range

* Although space is directly related to the sensor size, it can be extended by using a wide lens such as a fish-eye lens.
† Storage memory size.

1.2 Image Sensor Output and Structure of Image Signal

In this section, the structure of an image signal captured by image sensors is discussed.

1.2.1 Monochrome Still Images

As discussed in Section 1.1, the only image information required for monochrome still images is light intensity and space (position). The basic device configuration of image sensors is shown in Figure 1.4. Image sensors have an *image area* on which optical images are focused and are converted to image signals for output. In the image area, unit cells called pixels are arranged in a matrix in a plane. Each pixel has a sensor part typified by a photodiode, which absorbs incident light to generate a certain quantity of signal charges according to the light intensity. Thus, the light intensity information for a pixel is obtained at each sensor part. Figure 1.5b shows a partially enlarged image of Figure 1.5a, and the image signal of the same area captured by image sensors is shown in Figure 1.5c. The density of each rectangular block is the output of each pixel and the light intensity information $S(x_i, y_j)$ at the coordinate point (x_i, y_j) of each pixel. Expressing a two-dimensional coordinate point (x_i, y_j) using r_k , the signal can be written as $S(r_k)$. A set of $S(r_k)$ at all r_k in the image area makes up the information for one monochrome still image.

Dividing the image area into pixels that have a finite area, as shown in Figure 1.6, means fixing the area size and coordinate point at which light intensity information is picked up. The figure also shows that the continuous analog quantity, representing the position information, is replaced by discrete coordinate points. That is, x_i and y_j cannot take arbitrary values, but are built-in coordinate points that are fixed in imaging systems, which are the digitization of space coordinates. Using digitization in this system, since two-dimensional space information can be treated as determinate coordinate points, only light intensity information should be obtained. Therefore, the three-dimensional information of a monochrome still image is compressed into one dimension. Since the overall total

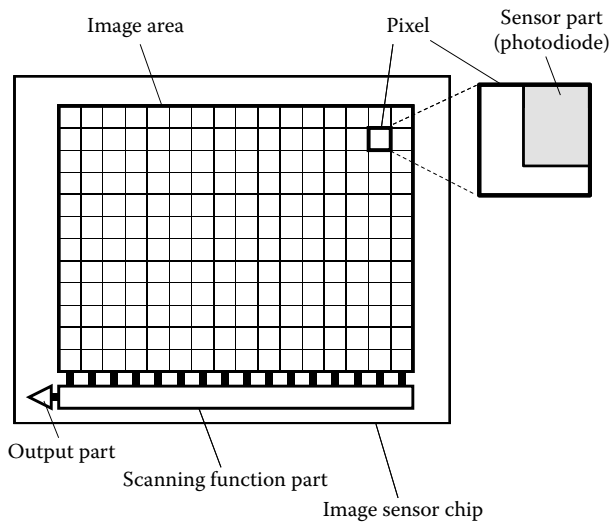


FIGURE 1.4
Basic device configuration of image sensors.

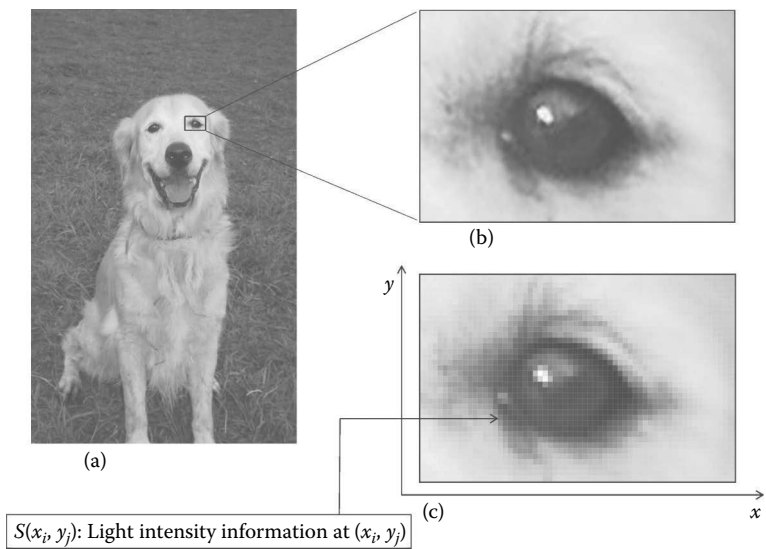


FIGURE 1.5 (See color insert) Optical image and the image signal obtained by image sensors: (a) optical image; (b) enlarged optical image; (c) image signal obtained by sensors.

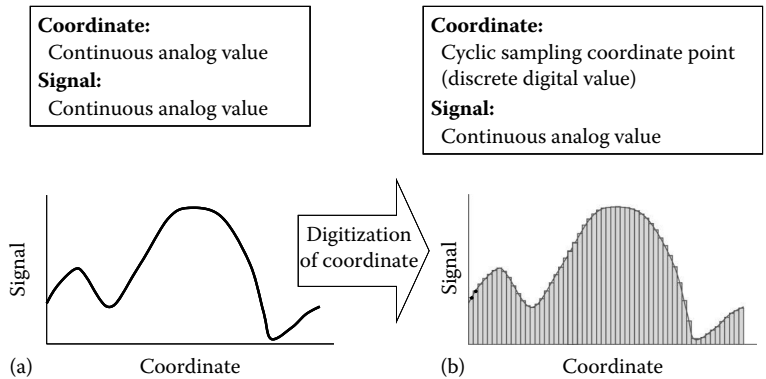


FIGURE 1.6 Digitization of coordinate: (a) space coordinate of an optical image; (b) space coordinate in an image sensor.

number of coordinate points is a pixel count of the image sensors, a larger number of pixels can achieve a higher space frequency, that is, a higher space resolution.

Image sensors also have a scanning function, transmitting the light intensity information of each pixel to the output part, which reads out the intensity information as an electric signal, as shown in Figure 1.4.

1.2.2 Color Still Images

The image information of color still images is formed by light intensity, space, and wavelength, as discussed in Section 1.1. Therefore, it is necessary to take the wavelength information in addition to the monochrome still image information. Although there are several ways to gather wavelength information, the most common method, the single-sensor color

system in which only one image sensor is used, is described. In this system, a color filter array (CFA) is arranged on a sensor. Figure 1.7 shows the Bayer CFA arrangement,⁷ which is by far the most commonly used CFA, with a pitch formed by 2×2 pixels. The pitch is composed of the three primary colors: red (R), green (G), and blue (B). On each pixel, only one color part of the filter is arranged. That is, one pixel for color imaging is formed from each one-color filter and the sensing part is formed under it. Since the human eye has the highest sensitivity to green, the G filter is arrayed in a checkerboard pattern* to contribute most to the space resolution. Red and blue are arranged between green on every two lines.

Using this configuration, each pixel contains information from only one color among R, G, and B. Although the wavelength distribution is continuous and analog, color information is restricted to only three types. Since the identification of a pixel also means the identification of its color, wavelength information also contains built-in coordinate points of color in the system as well as space information. By showing a coordinate point (R, G, or B) in three-dimensional space as c , the output from pixel k and color l can be expressed as $S(r_k, c_l)$. A set of signals from each pixel and color coordinate constructs the color still image information.

As a pixel and its color are specified at the same time in a single-sensor color system, k and l are immediately identified. The information on the physical quantity “wavelength” is replaced by perception “color.” In this system, the perceptual mechanism of color by the human eye and brain is availed of, and will be discussed in detail in Section 6.4.

1.2.3 Color Moving Images

The image information of color moving images requires all four factors (light intensity, space, wavelength, and time), as previously discussed. Almost all moving images are achieved by the continuous repetitive capturing and reproducing of still pictures, as shown in Figure 1.8. Thus, the basis of moving images is still images. Despite the name “still” images, their light intensity information is not that of an infinitesimal time length, but integrated information gathered during an exposure period of a certain length. This operation mode is called the integration mode. In this way, the signal amount can be increased by integrating the signal charges being generated throughout the exposure period. Since sensitivity can be drastically improved, almost all image sensors employ the integration mode.[†]

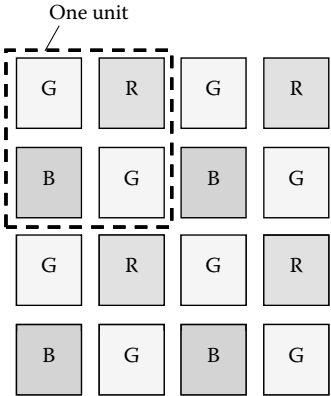
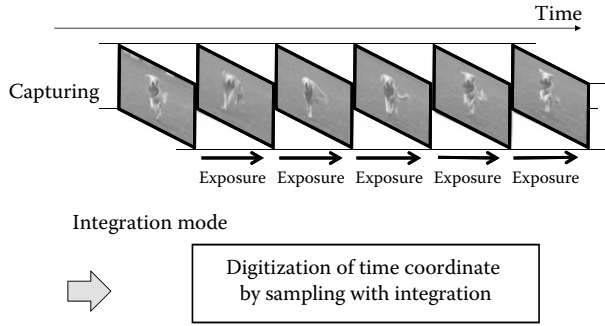


FIGURE 1.7
Example of the digitization of wavelength information: Bayer color filter array arrangement.

* The relationship between the arrays will be discussed in Section 7.2.
† “Almost all image sensors” means the sensors with integration mode.

**FIGURE 1.8**

Capturing the sequence of moving pictures.

In capturing moving images, still images are taken at a constant time interval. Although the physical quantity “time” has an essentially continuous analog distribution, images are picked up using timings and lengths determined by imaging systems. This is the digitization of the time coordinate. The output of pixel r_k , color c_l , and time t_f^* is $S(r_k, c_l, t_f)$, and a set of S makes up one moving picture. As a result of the formation of built-in six-dimensional coordinate points, the information that is to be captured, which is essentially seven dimensional, is compressed to one dimension of light intensity information only. It is also a significant data compression.

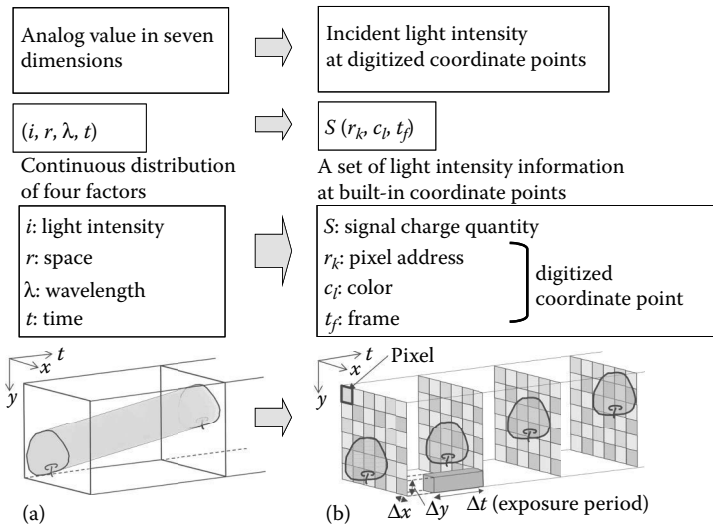
As was discussed, on one hand, the information that constructs optical images is a set of continuous analog quantities (intensity, space, wavelength, and time), as shown Figure 1.9a. On the other hand, an image signal captured by image sensors is only the light intensity information arriving at the built-in coordinate points of space, wavelength, and time, as shown in Figure 1.9b. This is how image sensors work. The entity of the built-in coordinate points is each pixel covered with a one-color filter having a corresponding spectral distribution during the exposure period, as shown in the graphics in Figure 1.9b. A pixel output S_{mn} at address r_{mn} is expressed by

$$S_{mn} = \iiint_{\Delta x \Delta y, \Delta \lambda, \Delta t} i(r, \lambda, t) f(r, \lambda) A(r, \lambda) dr d\lambda dt \quad (1.1)$$

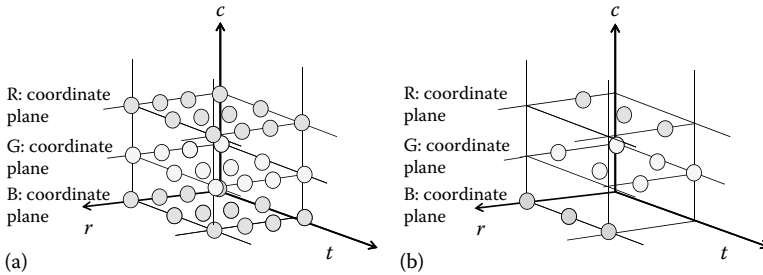
where i , f , and A are the light intensity distribution of the optical image, the spectral distribution of the color filter, and the spectral sensitivity distribution of the image sensor, respectively. By integrating at the ranges of $\Delta x \Delta y$ in space, $\Delta \lambda$ in the wavelength coordinate and Δt in time, the light intensity signal is sampled. Therefore, the quality of the information on space, color, and time is decided by the sampling frequency and the width of the built-in coordinate point in (r, c, t) space. That is, it is determined in the wake of the system design. What remains is the accuracy and the dynamic range of the light intensity signal S .

The general coordinate points of space, wavelength, and time digitized in (r, c, t) space are shown in Figure 1.10a. The position coordinate r , color coordinate c , and time coordinate t are the pixel at that position, the color of the filter at the pixel, and the frame at the number of the exposure in chronological order, respectively. The number of the coordinate point of the space coordinate r , the color coordinate, and time coordinate is the same as the

* As will be discussed in Chapter 4, a single exposure time in moving pictures is called a frame.

**FIGURE 1.9**

Comparison of optical image distribution with the image signal captured by image sensors: (a) optical image; (b) image sensor signal.

**FIGURE 1.10**

Digitized coordinate points: (a) digitized general coordinate points in a color system with an RGB color filter; (b) digitized coordinate points in a single-sensor color system with an RGB color filter.

pixel number in the image sensors, three in the case of the three primary colors system using RGB and the number of frames, three in the case of the three primary colors system using RGB and the number of frames, respectively. Since the color signal at the coordinate point r_k contains only one color information of three in a single-sensor color imaging system, only one-third of the information is captured in (r, c, t) space, as shown in Figure 1.10b. As the three color signals of RGB at each pixel are necessary to reproduce a color image, the two color signals missing in the system are estimated by signal processing such as demosaicking (color interpolation) using the correlation with the measured signal, as will be discussed in Section 8.2.3.

While the signal of almost all image sensors is light intensity information at digitized coordinate points, as shown in Section 1.2.3 by $S(r_k, c_l, t_f)$, different types of digitized coordinate points are possible. For example, as will be seen in Sections 5.3.3.2.3 and 7.3.2, there are sensors whose signal is time T , at which the change of light intensity information reaches a predetermined amount as $T(f(i), r_k)$, where $f(i)$ is a function of light intensity i .

1.3 Functional Elements of Image Sensors

Section 1.2 showed that the output of an image sensor is the intensity information of the incident light to each coordinate point. What functions must image sensors possess? First, they must have the function to measure the photon number, that is, the light intensity that comes to each coordinate point. Although, if it were possible,* an ideal method would be to count the photon numbers directly one by one; this is not realistic using the current technology. However, it can be achieved by using the following two steps. Initially, photons arriving at each coordinate point are absorbed to create signal charges (photoelectric conversion) by image sensors and the generated signal charges are integrated and stored (charge storage). In the second step, the stored charge quantity is measured and converted to electric signals (charge quantity measurement).

Secondary sensors must have the function to identify the signals from which coordinate point they emanate. Thus, image sensors must possess three functional elements: (1) a sensor part that generates and stores the signal charges; (2) a scanning part that identifies or addresses the coordinate point of the pixel of the signal; and (3) a signal charge quantity measuring part that measures the signal charge quantity and converts it to electric signals such as voltage, current, frequency, and pulse width.

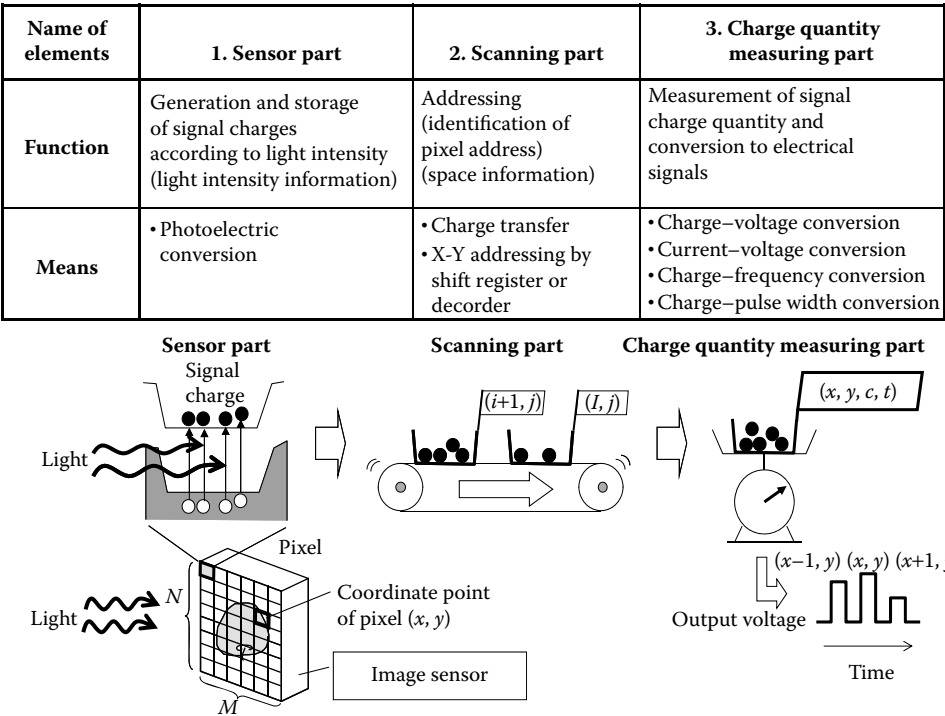


FIGURE 1.11
Functional elements of image sensors.

* It is possible to count the photon number directly only in the case of very low light intensity, when each photon comes with a sufficient time interval.

These elements are listed with their functions and material means in the table in Figure 1.11. The type of image sensor depends on the selection of the functional elements and their combination. The graphics in Figure 1.11 conceptually show the process by which the quantity information of the signal charges photoelectrically generated by the incident light to each pixel of an image sensor having $M \times N$ pixel number is transmitted with the address information and converted to electrical signals.

References

1. T. Kuroda, The 4 dimensions of noise, in *ISSCC 2007 Imaging Forum: Noise in Imaging Systems*, pp. 1–33, February 11–15, San Francisco, CA, 2007.
2. V. Gruev, A. Ortu, N. Lazarus, J. Spiegel, N. Engheta, Fabrication of a dual-tier thin film micro-polarization array, *Optics Express*, 15(8), 4994–5007, 2007.
3. T. Spirig, P. Seitz, O. Vietze, F. Heitger, The lock-in CCD-two-dimensional synchronous detection of light, *IEEE Journal of Quantum Electronics*, 31(9), 1705–1708, 1995.
4. H. Yabe, M. Ikeda, CMOS image sensor for 3-D range map acquisition using time encoded 2-D structured pattern, in *Proceedings of the 2011 International Image Sensor Workshop (IISW)*, p. 25, June 8–11, Hokkaido, Japan, 2011.
5. P. Seitz, Quantum-noise limited distance resolution of optical range imaging techniques, *IEEE Transactions on Circuits and Systems—I: Regular Papers*, 55(8), 2368–2377, 2008.
6. A. Wang, P.R. Gill, A. Molnar, An angle-sensitive CMOS imager for single-sensor 3D photography, in *Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, pp. 412–414, February 20–24, San Francisco, CA, 2011.
7. B.E. Bayer, Color imaging array, U.S. Patent 3,971,065, filed March 5, 1975, and issued July 20, 1976.

Task of Imaging and Role of Image Sensors

- T. Kuroda , The 4 dimensions of noise, in ISSCC 2007 Imaging Forum: Noise in Imaging Systems, pp. 1–33, February 11–15, San Francisco, CA, 2007.
- V. Gruev , A. Ortu , N. Lazarus , J. Spiegel , N. Engheta , Fabrication of a dual-tier thin film micro-polarization array, Optics Express, 15(8), 4994–5007, 2007.
- T. Spirig , P. Seitz , O. Vietze , F. Heitger , The lock-in CCD-two-dimensional synchronous detection of light, IEEE Journal of Quantum Electronics, 31(9), 1705–1708, 1995.
- H. Yabe , M. Ikeda , CMOS image sensor for 3-D range map acquisition using time encoded 2-D structured pattern, in Proceedings of the 2011 International Image Sensor Workshop (IISW), p. 25, June 8–11, Hokkaido, Japan, 2011.
- P. Seitz , Quantum-noise limited distance resolution of optical range imaging techniques, IEEE Transactions on Circuits and Systems—I: Regular Papers, 55(8), 2368–2377, 2008.
- A. Wang , P.R. Gill , A. Molnar , An angle-sensitive CMOS imager for single-sensor 3D photography, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC), pp. 412–414, February 20–24, San Francisco, CA, 2011.
- B.E. Bayer , Color imaging array, U.S. Patent 3,971,065, filed March 5, 1975, and issued July 20, 1976.

Device Elements and Circuits for Image Sensors

- W. F. Kosonocky , J. E. Carnes , Basic concept of charge-coupled devices, RCA Review, 38, 566–593, 1975.
- N. Koike , I. Takemoto , K. Sato , A. Sasano , S. Nagahara , M. Kubo , Characteristics of an *npn* structure MOS imager for color camera, Journal of the Institute of Television Engineers of Japan, 33(7), 548–553, 1978.
- W. F. Kosonocky , J. G. Carne , Charge-coupled digital circuit, in Proceedings of the IEEE International Solid-State Circuits Conference, Digest of Technical Papers, pp. 162–163, February, 1971.
- M. White , D. Lampe , F. Blaha , I. Mack , Characterization of surface channel CCD image arrays at low light levels, IEEE Journal of Solid-State Circuits, 9(1), 1–12, 1974.

Major Types of Noise in Image Sensors

- J.E. Carnes , W.F. Kosonocky , Noise source in charge-coupled devices, RCA Review, 33, 327–343, 1972.
- T. Obara , A. Yonezawa , A. Teramoto , R. Kuroda , S. Sugawa , T. Ohmi , Extraction of time constants ratio over nine orders of magnitude for understanding random telegraph noise in metal-oxide-semiconductor field-effect transistors, Japanese Journal of Applied Physics, 53, 04EC19, 2014.
- R. Kuroda , A. Yonezawa , A. Teramoto , T.-L. Li , Y. Tochigi , S. Sugawa , A statistical evaluation of random telegraph noise of in-pixel source follower equivalent surface and buried-channel transistors, IEEE Transactions on Electron Devices, 60(10), 3555–3561, 2013.

Types of Image Sensors

- T. Kuroda , Japan in the world of image sensors—What Japan has been playing the role of, and what could be learnt from the history, Journal of ITE, 65(3), 336–341, 2011.
- https://www.jstage.jst.go.jp/article/itej/65/3/65_3_336/_pdf (accessed January 10, 2014).
- W. S. Boyle , G. E. Smith , Charge coupled semiconductor devices, Bell System Technical Journal, 49, 587–593, 1970.
- R. H. Walden , R. H. Krambeck , R. J. Strain , J. McKenma , N. L. Schryer , G. E. Smith , The buried channel charge coupled device, Bell System Technical Journal, 51, 1635–1640, 1972.
- M. F. Tompsett , G. F. Amelio , W. J. Bertran , R. R. Buckley , W. J. Mcnamara , J. C. Mikkelsen , D. A. Sealer , Charge-coupled imaging devices: Experimental results, Transactions on Electron Device, 18(11), 992–996, 1971.
- G. F. Amelio , Physics and applications of charge-coupled devices, IEEE Intercon, 1/3, 1973.
- S. Okamoto , T. Kuroda , Japan patent, 3303384 (May 10, 2002).
- Y. Sone , K. Ishikawa , S. Hashimoto , T. Kuroda , Y. Ohkubo , A single chip CCD color camera system using field integration mode, Journal of the Institute of Television Engineers of Japan, 37(10), 855–862, 1983.
- Y. Ishihara , K. Tanigaki , A high photosensitivity IL-CCD image sensor with monolithic resin lens array, in Proceedings of the International Electron Devices Meeting, 19-3, pp. 497–500, 1983, Washington, DC.

A. Tsukamoto , W. Kamisaka , H. Senda , H. Niisoe , H. Aoki , T. Otagaki , Y. Shigeta , et al., High sensitivity pixel technology for a 1/4-inch PAL 430 k pixel IT-CCD, in Proceedings of the IEEE Custom Integrated Circuits Conference, pp. 39–42. 5–8 May 1996, San Diego, CA.

C. H. Sequin , Blooming suppression in charge coupled area imaging, Bell System Technical Journal, 51, 1923–1926, 1972. <http://www.alcatel.hu/bstj/vol51-1972/articles/bstj51-8-1923.pdf> (accessed January 10, 2014).

Y. Ishihara , E. Oda , H. Tanigawa , A. Kohno , N. Teranishi , E. Takeuchi , I. Akiyama , T. Kamata , Interline CCD image sensor with an anti blooming structure, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, pp. 168–169. 10–12 February 1982, San Francisco, CA.

T. Kuroda , K. Horii , S. Matsumoto , Y. Hiroshima , High performance interline transfer CCD, National Technical Report, 28(2), 12: 107–116, 1982.

N. Teranishi , A. Kohno , Y. Ishihara , E. Oda , K. Arai , No image lag photodiode structure in the interline CCD image sensor, IEDM Digest of Technical Papers, 12(6), 324–327, 1982.

T. Kuroda , K. Horii , Japan patent , 1996064 (March 13, 1992).

T. Kozono , T. Kuroda , Y. Matsuda , T. Yamaguchi , T. Kuriyama , S. Matsumoto , E. Fujii , Y. Hiroshima , K. Horii , Small image size CCD sensor, ITE Technical Report, TEBS 109-5, pp. 25–30, 1986.

B. C. Burkey , W. C. Chang , J. Littlehale , T. H. Lee , T. J. Tredwell , J. P. Lavine , E. A. Trabka , The pinned photodiode for an interline-transfer CCD image sensor, IEDM Digest of Technical Papers, 2(3), 28–31, 1984.

K. Horii , T. Kuroda , T. Kunii , A new configuration of CCD imager with a very low smear level, Electron Device Letters, EDL-2, 12, 1981.

G. P. Weckler , A silicon photodevice to operate in a photon flux integration mode, in Proceedings of International Electron Devices Meeting Digest of Technical Papers, 10-5, October 1965, Washington, DC.

G. P. Weckler , Operation of p-n junction photodetectors in a photon flux integrating mode, Journal of Solid-State Circuits, 2, 65–73, 1967.

M. Masuda , M. Noda , Y. Saito , S. Ohba , K. Takahashi , T. Fujita , Solid state color camera with single-chip MOS imager, ITE Technical Report, 4, 41, TEBC69-1, pp. 1–6, February 1981.

I. Takemoto , MOS imager, Journal of ITE, 40(11), 1067–1072, 1986.

M. Aoki , H. Ando , S. Ohba , I. Takemoto , S. Nagahara , T. Nakano , M. Kubo , T. Fujita , 2/3" inch format MOS single-chip color imager, IEEE Transactions on Electron Devices, ED-29(4), 745–750, 1982.

K. Takahashi , N. Ozawa , M. Aoki , I. Takemoto , T. Suzuki , T. Miyazawa , S. Nagahara , High resolution MOS area imaging device, ITE Technical Report, 9, 17, ED-623, pp. 13–18, February 1982.

K. Takahashi , S. Nagahara , I. Takemoto , M. Aoki , N. Ozawa , T. Suzuki , High resolution MOS area sensor, Journal of the Institute of Television Engineers of Japan, 37(10), 812–818, 1983.

I. Takemoto , T. Miyazawa , S. Nishizawa , M. Uehara , M. Nakai , T. Akiyama , TSL (transversal signal line) solid state imager, ITE Technical Report, 9, 17, ED-891, pp. 49–54, September 1985.

P. Noble , Self-scanned silicon image detector arrays, Transactions on Electron Devices, ED-15, 202–209, 1968.

F. Andoh , J. Matsuzaki , An imaging device with amplifier in each photo sensing element, in Proceeding of IEICE General Conference, No. 1159, 1981, Tokyo.

F. Andoh , K. Taketoshi , J. Yamazaki , M. Sugawara , Y. Fujita , K. Mitani , Y. Matuzawa , K. Miyata , S. Araki , A 250000-pixel image sensor with FET amplification at each pixel for high-speed television cameras, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, pp. 212–213. 14–16 February 1990, San Francisco, CA.

A. Theuvsen , 50 years of solid-state image sensors at ISSCC, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, S26, 2003, San Francisco, CA.

N. Tanaka , S. Hashimoto , M. Shinohara , S. Sugawa , M. Morishita , S. Matsumoto , Y. Nakamura , T. Ohmi , A 310 k pixel bipolar imager (BASIS), in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, WPM 8.5, pp. 96–97, 1989, New York.

J. Hynecek , A new device architecture suitable for high-resolution and high-performance image sensors, Transactions on Electron Devices, 35(5), 646–652, 1988.

E. Oba , K. Mabuchi , Y. Lida , N. Nakamura , H. Miura , A 1/4 inch 330 k square pixel progressive scan CMOS active pixel image sensor, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, pp. 180–181, 1997, San Francisco, CA.

Y. Matsunaga , Y. Endo , Noise cancel circuit for CMOS image sensor, ITE Technical Report, 22(3), 7–11, 1998.

M. Mori , M. Katsuno , S. Kasuga , T. Murata , T. Yamaguchi , A 1/4in 2M pixel CMOS image sensor with 1.75 transistor/pixel, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 6.2, pp. 80–81, 2004, San Francisco, CA.

K. Mabuchi , N. Nakamura , E. Funatsu , T. Abe , T. Umeda , T. Hoshino , R. Suzuki , H. Sumi , CMOS image sensor using a floating diffusion driving buried photodiode, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 6.3, pp. 82–83, 2004, San Francisco, CA.

H. Takahashi , M. Kinoshita , K. Morita , T. Shirai , T. Sato , T. Kimura , H. Yuzurihara , S. Inoue , A 3.9 μ m pixel pitch VGA format 10b digital image sensor with 1.5-transistor/pixel, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 6.1, pp. 108–109, 2004, San

San Francisco, CA.

K. Itonaga , K. Mizuta , T. Kataoka , M. Yanagita , H. Ikeda , H. Ishiwata , Y. Tanaka , et al., Extremely-low-noise CMOS image sensor with high saturation capacity, in Proceedings of the IEEE International Electron Devices Meeting Digest of Technical Papers, 8.1.1–8.1.4, pp. 171–174. 5–7 December 2011, Washington, DC.

S. Kawahito , M. Sakakibara , D. Handoko , N. Nakamura , H. Satoh , M. Higashi , K. Mabuchi , H. Sumi , A column-based pixel-gain-adaptive CMOS image sensor for low-light-level imaging, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 12.7, pp. 224–225, 2003, San Francisco, CA.

H. Takahashi , T. Noda , T. Matsuda , T. Watanabe , M. Shinohara , T. Endo , S. Takimoto , et al., A 1/2.7 inch low-noise CMOS image sensor for full HD camcorders, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 28.6, pp. 510–511. 11–15 February 2007, San Francisco, CA.

B. Pain , E. Fossum, Approaches and analysis for on-focal-plane analog-to-digital conversion, in Proceedings of the SPIE, Volume 2226, Aerospace Sensing—Infrared Readout Electronics II, pp. 1–11, 1994, San Diego, CA.

B. Fowler , D. Yang , A. E. Gamal , Techniques for pixel level analog to digital conversion , Aerosense, 98, 3360–3361, 1998. http://www-isl.stanford.edu/~abbas/group/papers_and_pub/aerosense98_slide.pdf (accessed January 10, 2014).

F. Andoh , M. Nakayaka , H. Shimamoto , Y. Fujita , A digital pixel image sensor with 1-bit ADC and 8-bit pulse counter in each pixel, IISW, P1, (1999).
<http://www.imagesensors.org/Past%20Workshops/1999%20Workshop/1999%20Papers/12%20Andoh%20et%20al.pdf> (accessed January 10, 2014).

K. Cho , C. Lee , S. Eikedal , A. Baum , J. Jiang , C. Xul , X. Fan , R. Kauffman , A 1/2.5 inch 8.1 Mpixel CMOS image sensor for digital cameras, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 28.5, pp. 508–509, 2007, San Francisco, CA.

Y. Nitta , Y. Muramatsu , K. Amano , T. Toyama , J. Yamamoto , K. Mishina , A. Suzuki , et al., High-speed digital double sampling with analog CDS on column parallel ADC architecture for low-noise active pixel sensor, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 27.5, pp. 2024–2031. 6–9 February 2006, San Francisco, CA.

S. Yoshihara , M. Kikuchi , Y. Ito , Y. Inada , S. Kuramochi , H. Wakabayashi , M. Okano , et al., A 1/1.8-inch 6.4 M Pixel 60 frames/s CMOS image sensor with seamless mode change, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 27.1, pp. 1984–1993. 6–9 February 2006, San Francisco, CA.

S. Matsuo , T. J. Bales , M. Shoda , S. Osawa , K. Kawamura , A. Andersson , M. Haque , et al., 8.9-megapixel video image sensor with 14-b column-parallel SA-ADC, Transactions on Electron Devices, 56(11), 2380–2389, 2009.

J. Park , S. Aoyama , T. Watanabe , K. Isobe , S. Kawahito , A high-speed low-noise CMOS image sensor with 13-b column-parallel single-ended cyclic ADCs, Transactions on Electron Devices, 56(11), 2414–2422, 2009.

J. Park , S. Aoyama , T. Watanabe , T. Akahori , T. Kosugi , K. Isobe , Y. Kaneko , et al., A 0.1 μ m vertical FPN 4.7 μ s read noise 71 dB DR CMOS image sensor with 13b column-parallel single-ended cyclic ADCs, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 15.3, pp. 268–269. 8–12 February 2009, San Francisco, CA.

Y. Chae , J. Cheon , S. Lim , D. Lee , M. Kwon , K. Yoo , W. Jung , D. Lee , S. Ham , G. Han , A 2.1 M pixel 120 frame/s CMOS image sensor with column-parallel $\Delta\Sigma$ ADC architecture, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 22.1, pp. 394–395. 7–11 February 2010, San Francisco, CA.

Y. Chae , J. Cheon , S. Lim , M. Kwon , K. Yoo , W. Jung , D. Lee , S. Ham , G. Han , A 2.1 M pixels, 120 frame/s CMOS image sensor with column-parallel 16 ADC architecture, Journal of Solid State Circuits, 46(1), 236–247, 2011.

W. Yang , A wide-dynamic-range, low-power photosensor array, in Proceedings of the IEEE International Solid-State Circuits Conference, 41st ISSCC Digest of Technical Papers, 13.7, pp. 230–231. 16–18 February 1994, San Francisco, CA.

J. Gambino , B. Leidy , J. Adkisson , M. Jaffe , R. J. Rassel , J. Wynne , J. Ellis-Monaghan , et al., CMOS imager with copper wiring and lightpipe, in Proceedings of the International Electron Devices Meeting, Technical Digest, pp. 5.5.1–5.5.4. 11–13 December 2006, San Francisco, CA.

J. Tower , P. Swain , F. Hsueh , R. Dawson , P. Levine , G. Meray , J. Andrews , et al., Large format backside illuminated CCD imager for space surveillance, Transactions on Electron Devices, 50(1), 218–224, 2003.

S. Iwabuchi , Y. Maruyama , Y. Ohgishi , M. Muramatsu , N. Karasawa , T. Hirayama , A back-illuminated high-sensitivity small-pixel color CMOS image sensor with flexible layout of metal wiring, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 16.8, pp. 1171–1178. 6–9 February 2006, San Francisco, CA.

Papers in IISW 2009, Backside Illumination Symposium , IISW (June 2009). <http://www.imagesensors.org/Past%20Workshops/2009%20Workshop/2009%20Papers/2009%20IISW%20Program.htm> (accessed January 10, 2014).

M. Nakai , K. Watanabe , I. Takemoto , S. Shimada , T. Nagano , Si substrate for low noise color image sensor, in Proceeding of ITE General Conference, pp. 21–22, 1982, Tokyo.

Y. Hiroshima , T. Kuroda , S. Matsumoto , K. Horii , T. Kunii , CCD image sensor fabricated on EPI-wafers, IEICE Technical Report, SSD81-129, pp. 103–108, February 1982.

S. Sukeyawa , T. Umebayashi , T. Nakajima , H. Kawanobe , K. Koseki , I. Hirota , T. Haruta , et al., A 1/4-inch 8 M pixel back-illuminated stacked CMOS image sensor, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 27.4, pp. 484–486, 2013, San Francisco, CA.

H. Watanabe , J. Hirai , M. Katsuno , K. Tachikawa , S. Tsuji , M. Kataoka , S. Kawagishi , et al., A 1.4 μm front-side illuminated image sensor with novel light guiding structure consisting of stacked lightpipes, in Proceedings of the IEEE International Electron Devices Meeting, Technical Digest, 8.3, pp. 179–182. 5–7 December 2011, Washington, DC.

N. Teranishi , H. Watanabe , T. Ueda , N. Sengoku , Evolution of optical structure in image sensors, in Proceedings of the IEEE International Electron Devices Meeting, Technical Digest, 24.1, pp. 533–536. 10–13 December 2012, San Francisco, CA.

T. Kuroda , M. Masuyama , U.S. Patent 6469740.

M. Ihama , H. Inomata , H. Asano , S. Imai , T. Mitsui , Y. Imada , M. Hayashi , et al., CMOS image sensor with an overlaid organic photoelectric conversion layer, IISW, P33, 2011.

S. Isono , T. Satake , T. Hyakushima , K. Taki , R. Sakaida , S. Kishimura , S. Hirao , et al., A 0.9 μm pixel size image sensor realized by introducing organic photoconductive film into the BEOL process, in Proceedings of the 2013 IEEE International Interconnect Technology Conference, paper ID 3030, 2013, Hokkaido, Japan.

M. Mori , Y. Hirose , M. Segawa , I. Miyana , R. Miyagawa , T. Ueda , H. Nara , et al., Thin organic photoconductive film image sensors with extremely high saturation of 8500 electrons/ μm^2 , in 2013 Symposium on VLSI Technology, 2-4, pp. 22–24. 11–13 June 2013, Kyoto.

M. Ishii , S. Kasuga , K. Yazawa , Y. Sakata , T. Okino , Y. Sato , J. Hirase , et al., An ultra-low noise photoconductive film image sensor with a high-speed column feedback amplifier noise canceller, in 2013 Symposium on VLSI Circuits, 2-3, pp. 8–9. 12–14 June 2013, Kyoto.

K. Yasutomi , S. Itoh , S. Kawahito , A 2.7e—Temporal noise 99.7% shutter efficiency 92 dB dynamic range CMOS image sensor with dual global shutter pixels, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 22.3, pp. 398–399, 2010, San Francisco, CA.

M. Sakakibara , Y. Oike , T. Takatsuka , A. Kato , K. Honda , T. Taura , T. Machida , et al., An 83 dB-dynamic-range single-exposure global-shutter CMOS image sensor with in-pixel dual storage, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 22.1, pp. 380–382. 19–23 February 2012, San Francisco, CA.

Y. Tochigi , K. Hanzawa , Y. Kato , R. Kuroda , H. Mutoh , R. Hirose , H. Tominaga , K. Takubo , Y. Kondo , S. Sugawa , A global-shutter CMOS image sensor with readout speed of 1 T pixel/s burst and 780 M pixel/s continuous, in Proceedings of the IEEE International Solid-State Circuits Conference Digest of Technical Papers, 22.2, pp. 382–384. 19–23 February 2012.

S. G. Smith , J. E. D. Hurwitz , M. J. Torrie , D. J. Baxter , A. A. Murray , P. Likoudis , A. J. Holmes , et al., A single-chip CMOS 306 $\times$ 244-pixel NTSC video camera and a descendant coprocessor device, Journal of Solid State Circuits, 33(12), 2104–2111, 1998.

D. Renshaw , P. B. Denyer , G. Wang , M. Lu , ASIC vision, in Proceedings of the IEEE 1990 Custom Integrated Circuits Conference, pp. 7.3.1–7.3.4, May 1990, Boston, MA.

Impacts of Digitization by Built-In Coordinate Points on Image Information Quality

S. Nagahara , Y. Kobayashi , S. Nobutoki , T. Takagi , Development of a single pickup tube color television camera by frequency multiplexing, Journal of the Institute of Television Engineers of Japan, 26(2), 104–110, 1972.

S. Nagahara , http://www.ieice-hbkb.org/files/08/08gun_04hen_02-04.pdf, pp. 116–118, October 25, 2011 (accessed January 10, 2014).

B. Wandell , Foundations of color vision: Retina and brain, in ISSCC 2006 Imaging Forum—Color Imaging, pp. 1–23, February 9, San Francisco, CA, 2006, <http://white.stanford.edu/~brian/papers/ise/ISSCC-2006-Wandell-ColorForum.pdf> (accessed January 10, 2014).

Y. Sone , K. Ishikawa , S. Hashimoto , T. Kuroda , Y. Ohkubo , A single chip CCD color camera system using field integration mode, Journal of the Institute of Television Engineers of Japan, 37(10), 855–862, 1983.

Technologies to Improve Image Information Quality

- S.G. Chamberlain , J.P.Y. Lee , A novel wide dynamic range silicon photodetector and linear imaging array, *Transaction on Electron Devices*, ED-31(2), 175–182, 1984.
- H. Komobuchi , A. Fukumoto , T. Yamada , Y. Matsuda , T. Kuroda , 1/4 inch NTSC format hyper-D range IL-CCD, in *IEEE Workshop on CCDs and Advanced Image Sensors*, April 20–22, Dana Point, CA, 1995, <http://www.imagesensors.org/Past%20Workshops/1995%20Workshop/1995%20Papers/02%20Komobuchi%20et%20al.pdf> (accessed January 10, 2014).
- N. Akahane , S. Sugawa , S. Adachi , K. Mori , T. Ishiuchi , K. Mizobuchi , A sensitivity and linearity improvement of a 100 dB dynamic range CMOS image sensor using a lateral overflow integration capacitor, in *2005 Symposium on VLSI Circuits, Digest of Technical Papers*, pp. 62–65, June 16–18, Kyoto, Japan, 2005.
- N. Akahane , S. Sugawa , S. Adachi , K. Mori , T. Ishiuchi , K. Mizobuchi , A sensitivity and linearity improvement of a 100-dB dynamic range CMOS image sensor using a lateral overflow integration capacitor, *IEEE Journal of Solid-State Circuits*, 41(4), 851–856, 2006.
- T. Yamada , K. Ikeda , Y. Kim , H. Wakoh , T. Toma , T. Sakamoto , K. Ogawa , et al., A progressive scan CCD image sensor for DSC applications, *Journal of Solid-State Circuits*, 35(12), 2044–2054, 2000.
- A. Watanabe , T. Mori , S. Nagata , K. Hiwatashi , Spatial sine-wave responses of the human visual system, *Vision Research*, 8, 1245–1263, 1968.
- H. Miyahara , The picture quality improve technology for consumer video camera, *Journal of ITE*, 63(6), 731–734, 2009.
- Sony Corporation . <http://www.sony.jp/products/Consumer/handycam/PRODUCTS/special/02cmos.html> (accessed January 10, 2014).
- B. Cremers , M. Agarwal , T. Walschap , R. Singh , T. Geurts , A high speed pipelined snapshot CMOS image sensor with 6.4 Gpixel/s data rate, in *Proceedings of 2009 International Image Sensor Workshop*, p. 9, June 22–28, Bergen, Norway, 2009, http://www.imagesensors.org/Past%20Workshops/2009%20Workshop/2009%20Papers/030_paper_cremers_cypress_gs.pdf (accessed January 10, 2014).
- W. Kosonocky , G. Yang , C. Ye , R. Kabra , L. Xie , J. Lawrence , V. Mastrocolla , F. Shallcross , V. Patel , 360 × 360-element very-high frame-rate burst-image sensor, in *Proceedings of the IEEE Solid-State Circuits Conference Digest of Technical Papers*, 11.3, pp. 182–183, February 8–10, San Francisco, CA, 1996.
- T. Kuroda , Private communication at Kyoto Research Laboratory, Panasonic Corporation, April 1996.
- T. Etoh , D. Poggemann , A. Ruckelshausen , A. Theuwissen , G. Kreider , H. Folkerts , H. Mutoh , et al., A CCD image sensor of 1 Mframes/s for continuous image capturing of 103 frames, in *Proceedings of IEEE International Solid-State Circuits Conference Digest of Technical Papers*, 2.7, pp. 46–47, February 3–7, San Francisco, CA, 2002.
- T. Etoh , D. Nguyen , S. Dao , C. Vo , M. Tanaka , K. Takehara , T. Okinaka , et al., A 16 Mfps 165 kpixel backside-illuminated CCD, in *Proceedings of IEEE International Solid-State Circuits Conference Digest of Technical Papers*, 23.4, pp. 406–408, February 20–24, San Francisco, CA, 2011.
- Y. Tochigi , K. Hanzawa , Y. Kato , R. Kuroda , H. Mutoh , R. Hirose , H. Tominaga , K. Takubo , Y. Kondo , S. Sugawa , A global-shutter CMOS image sensor with readout speed of 1 Tpixel/s burst and 780 Mpixel/s continuous, in *Proceedings of IEEE International Solid-State Circuits Conference Digest of Technical Papers*, 22.2, pp. 382–384, February 19–23, San Francisco, CA, 2012.
- P. Lichtsteiner , C. Posch , T. Delbruck , A 128 × 128 120 dB 30 mW asynchronous vision sensor that responds to relative intensity change, in *Proceedings of IEEE International Solid-State Circuits Conference Digest of Technical Papers*, 27.9, pp. 508–510, February 6–9, San Francisco, CA, 2006.
- P. Lichtsteiner , C. Posch , T. Delbruck , A 128 × 128 120 dB 15 μ s latency asynchronous temporal contrast vision sensor, *Journal of Solid-State Circuits*, 43(2), 566–576, 2008.
- T. Delbruck , Dynamic vision sensor (DVS): Asynchronous temporal contrast silicon retina, sili-conretina, 2013. <http://siliconretina.ini.uzh.ch/wiki/index.php> (accessed January 10, 2014).
- Y. Yokoyama , T. Hasegawa , N. Tsumura , H. Haneishi , Y. Miyake , New color management system on human perception and its application to recording and reproduction of art (I)—Design of image acquisition system, *Journal of SPIE*, 61(6), 343–355, 1998.
- M. Yamaguchi , T. Teraji , K. Ohsawa , T. Uchiyama , H. Motomura , Y. Murakami , N. Ohyama , Color image reproduction based on the multispectral and multi-primary imaging: Experimental evaluation, *Proceedings of SPIE*, 4663, 15–26, 2002.
- Shin, Satori . http://www.nikko-pb.co.jp/nk_comm/mok08/html/images/1203g61.pdf (accessed January 10, 2014).

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T. Kuroda, The 4 dimensions of noise, IEEE International Solid-State Circuits Conference, San Francisco, February 2007, Imaging Forum: Noise in imaging systems, pp. 1–33.